MEDICAL INSTRUMENTATION LAB REPORT

Introduction

Electrocardiogram (ECG) is a medical device that is able to measure and record the electrical signal or impulses from the heart to detect and monitor any heart issues*[Mayo Clinic , 2022]*. ECG is carried out by attaching small, sticky sensor electrodes to the patient's arms, legs and chest with the electrode wires connected to an ECG recording machine and observed via a monitor*[nhs.uk, 2017]*.

Methods - Circuit

Whole Circuit Diagram

(see Appendix 1 for the whole circuit diagram)

Design Rationale

Voltage Divider

Normally, ECG testing uses a dual supply voltage where the range of the op-amps are operated between the positive and negative voltages. However, in this case, a single supply design is used as the Arduino Uno used can only handle a single supply voltage from a maximum of 5V to a minimum of 0V. The difference between the two designs is that dual supply supplies a positive and negative voltage source allowing the signal to oscillate between the positive and negative region while a single supply only supplies the positive source with the other being grounded allowing a half-oscillating signal between the positive region and ground(0V)*[www.learningaboutelectronics.com n.d]*. Hence a voltage divider is used to provide a virtual ground of 2.5V to allow a full sinusoidal wave for the subsequent circuits. An equal 1kΩ resistors are used to provide exactly half the input voltage of 2.5V.

$$\text{Vout} = \frac{1}{1+1} * 5\text{V} = 2.5\text{V}$$

Instrumentation Amplifier

Instrumentation amplifier is very widely used due to its ability to suppress unwanted commnon-mode signals as it performs a differential amplification*[Xiu and Li, 2012]*. A ready to use pre-built intrumentation amplifier is used instead of building it from scratch with op-amps is to reduce the time, breadboard space and resources required to build the whole circuit. In addition to that, it also consumes less power than a manually built version*[Xiu and Li, 2012]*. An INA126 is a precision component giving accurate, low-noise differential-signal acquisiiton having a wide voltage range between ± 1.35V and ± 18V*[Texas Instruments, 2022]*. A 10kΩ resistor is chosen as Rg to produce an intial gain of 13 using formula from the INA126 datasheet.

$$Gain = 5 + (\frac{80000}{Rg})$$

$$Gain = 5 + (\frac{80000}{10000}) = 13$$

Filter Specifications and Characterizations

There are a lot of noises and interference involved when using an ECG circuit like baseline wander noise and electromyographic (EMG) noise. Filters are used to remove such noises from disrupting the final measurement. The frequency of the baseline wander noise is around 0.5Hz so a high pass filter with a cut-off frequency of 0.5Hz is used to filter this noise out. The noise frequency of the EMG noise is more than a 100Hz, as a precaution a 2nd Order Sallen Key low pass filter with a cut-off frequency of 150Hz is used for filtering*[Kher, 2019][Christov et al, 2017]*. A 2nd order filter is chosen in this case due to it having a high input impedance and a low output impedance which protects the instrument providing the signal from excessive current that could cause a drop in the voltage measurement or damage to the instrument*[hades.mech.northwestern.edu, 2008]*.

High-Pass Filter

For the high pass filter, a capacitor of 4.7µF and a resistor of 68kΩ are used to reach an almost ideal 0.5Hz cut-off frequency. *(see Appendix for the tabled data)*

$$fc = \frac{1}{2\pi RC}$$

$$fc = \frac{1}{2\pi(68x10\text{^}3)(4.7x10\text{^}-6)} = 0498\ Hz$$

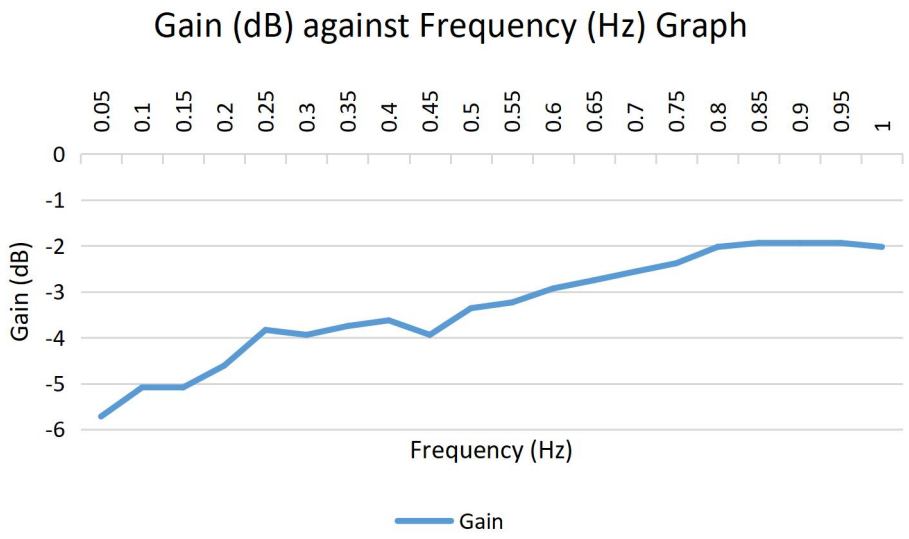


Figure 1 : Gain(dB) against Frequency(Hz) Graph of High Pass Filter

From the figure above, it is observed that the overall general shape of the graph closely follows that of an ideal high pass filter although being too linear with the absence of a slope. This may be that the filter was still connected to the output of the INA126 when measuring the oscilloscope. Not isolating the high pass filter may be the cause of this linear graph. This however does not mean that the filter is not working as shown by the graph above.

Sallen-Key Low Pass Filter

For the 2nd order low pass filter, capacitors of 1µF and resistors of 1kΩ was used to reach an almost ideal 150Hz cut-off frequency. *(see Appendix for tabled data)*

$$fc = \frac{1}{2\pi RC}$$

$$fc = \frac{1}{2\pi(1x10^{3})(1x10^{-6})} = 159 \text{ Hz}$$

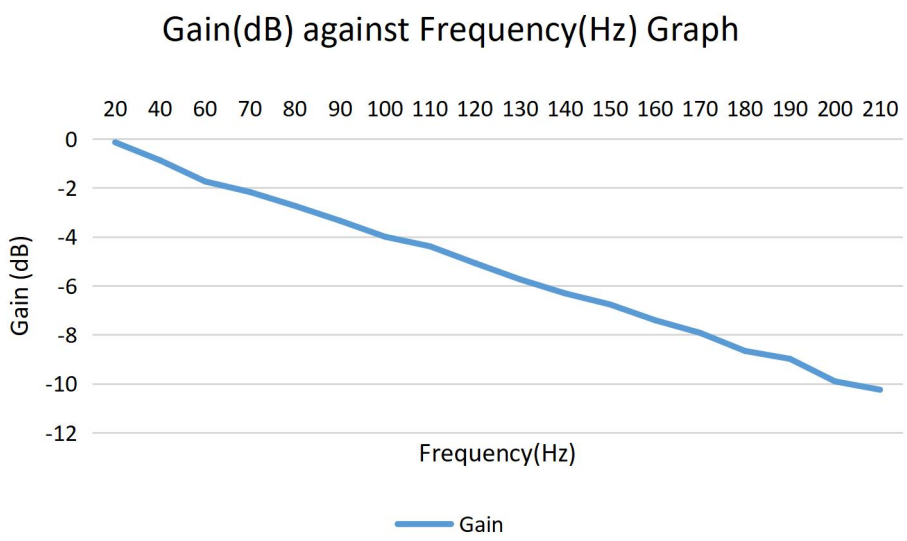


Figure 2 : Gain(dB) against Frequency(Hz) Graph of 2nd Order Sallen Key Low Pass Filter

Similarly, for the 2nd order low pass filter, the graph shows a decreasing linear pattern from what is expected of a slope in a low pass filter. As explained previously, this may be due to it not being isolated because it was connected into the output of the high pass filter and INA126 before it. The filter seems to be working fine despite this.

Non-Inverting Amplifier

A non-inverting amplifier is used as the final gain after the noise has been filtered. A gain of 34 is obtained using a 33kΩ and a 1kΩ resistor.

$$Gain = 1 + (\frac{R2}{R1})$$

$$Gain = 1 + (\frac{33}{1}) = 34$$

With the gain of the instrumentation amplifier being 13 and the non-invering amplifier being 34. The total gain of the whole circuit is (13 x 34 = 442). For a target muscle, the average input signal produced is between 1-10mV*[Zahak, 2012]*. So the maximum final output voltage that can be measured is (10x10^-3 x 442 = 4.42 V) which is within the range of the Arduino Uno. The whole gain is not implemented straight from the INA126 but divided between the INA126 and non-inverting amplifier is because when a signal is amplified so is the noise. Hence, a small gain is implemented initially followed by filtering systems and finally greatly amplified by the non-inverting amplifier.

Testing

Before the ECG was tested out using the electrodes provided, a test run was performed using the function generator and the oscilloscope to see whether a desired or reasonable outcome has been achieved.

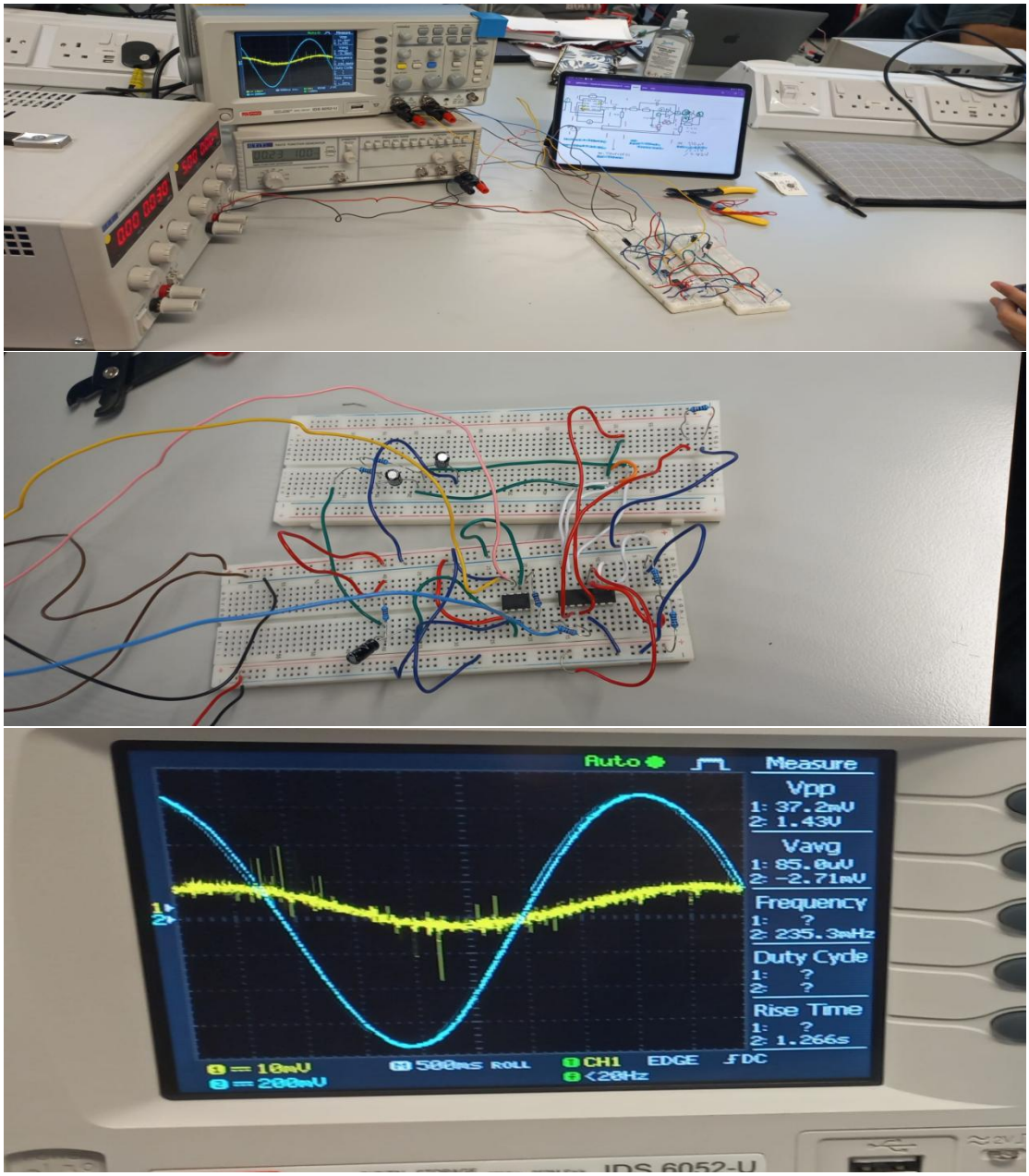


Image 1 : Images of the whole circuit set-up using function generator and oscilloscope

As observed on the oscilloscope, the input from the function generator is 37.2mV while the output of the circuit reaches 1.43mV signifying that the signal has been amplified to about 38.4 times the original input signal. Additionally, the output range is within the range of the Arduino Uno which is around 0V - 5V meaning that there will be a reading on the serial monitor and the serial plotter of the Arduino. Hence, the circuit is now feasible to be used with the Arduino Uno.

Methods - Program

Before testing out on a patient, the Arduino Uno has to be initially coded and uploaded into it via the Arduino IDE software. *(see coding for ECG testing in the Appendix).* Timer counters TCNT values is set to determine the sampling frequency of the ADC. The sampling frequency has to be twice that of the chosen frequency according to the Nyquist Theorem. The chosen sampling frequency is 250Hz so a 500

Hz is used based on the Nyquist Theorem application. *[Pizzati, Cifaldi and Nolfe, 1985]*. The value of TCNT value can be determined by the following calculations:

$$\frac{16MHz}{256\ prescaler} = 62500Hz$$

$$1\ bitstep = \frac{1}{62500} = 1.6\times10^{-5}s$$

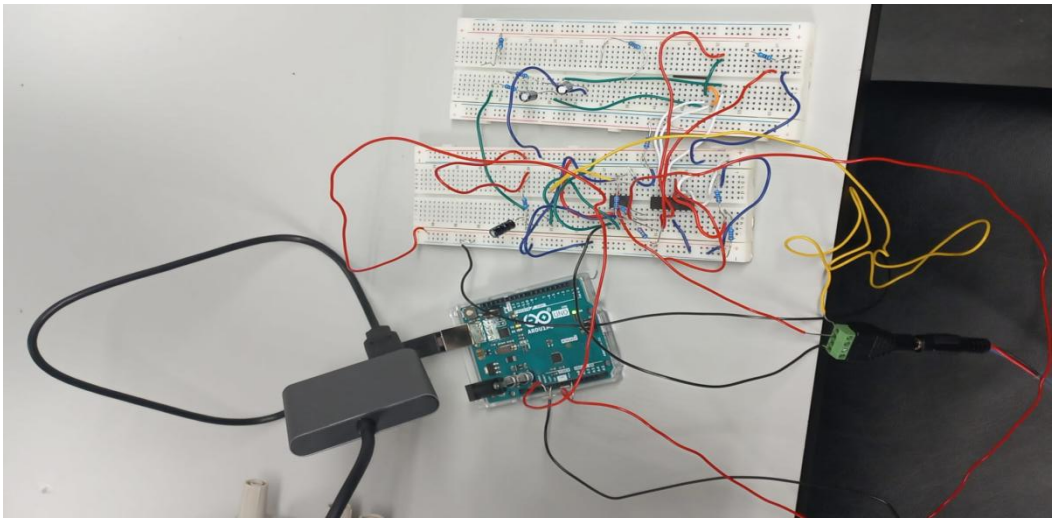
$$steps\ to\ overflow = \frac{2\times10^{-3}}{1.6\times10^{-5}} = 125\ steps$$

$$TCNT\ set\ value = (2\text{^}16 - 125) = 65411$$

The resolution is 1.6x10^-5 seconds after the clock frequency is reduced by the prescaler. This is done in order for the overflow interrupt code to happen within the range of 0 to 65535 in the Arduino. Note that the code in the interrupt has to be kept short and concise as the longer codes written, the command in the ISR routine takes longer to run and there are chances that other interrupts coded can disrupt or prevent other interrupts from running*[Staff, 2012]*.

Electrode Placements

Now that the coding and the circuit are both feasible, the experiment can now be run using real electrodes on a patient's body. For the 2 input electrodes are placed on the patient's left and right arm while the ground electrode is placed on the patient's right leg. A 3-electrode ECG is commonly used to record and monitor a 24-hour reading. It is frequently used tool for heart problem diagnoses*[www.cardiosecur.com, n.d]*. The reference electrode is connected to the right leg to reduce interference without affecting the ECG results because of the Einthoven's triangle electrode configuration*[Abi Saleh and Omar, 2010]*. The arm electrodes are placed on the "flexor carpi radialis" which is a surface muscles. The electrodes are placed on the belly of the muscle as this location gives the strongest muscle signal. Electrodes cannot be placed on the tendon or the edge of the muscles. Muscle fibres becomes weaker near the tendons giving smaller signals and cross-talk interference from adjacent muscles will increase when placed near the muscle edge*[Luca, 2002]*.



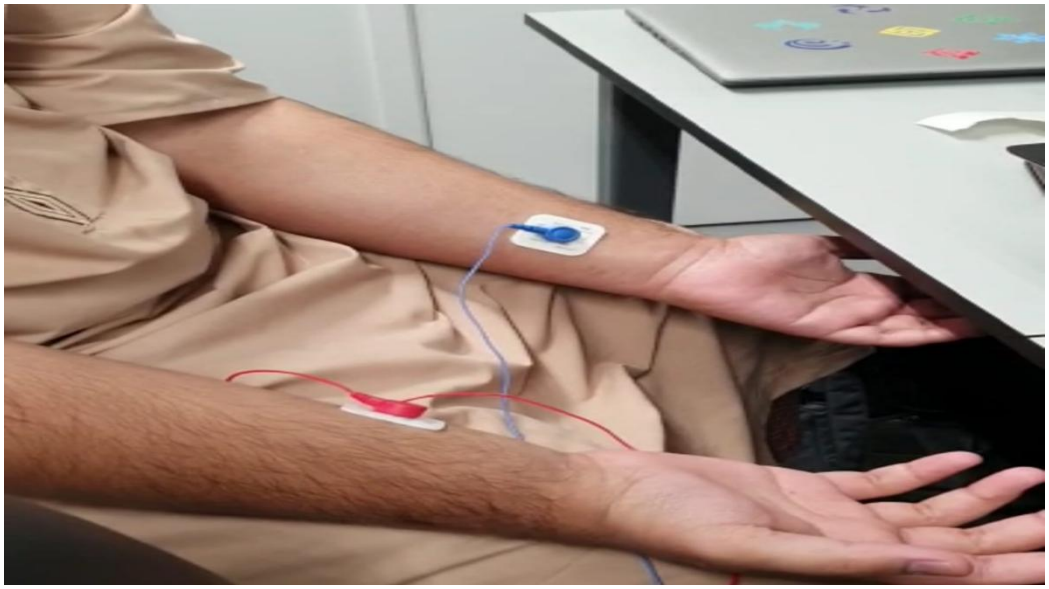


Image 2 : Circuit and Arduino Uno set up for electrodes on a patient's body(left and right arm, leg is hidden due to camera angle)

Results

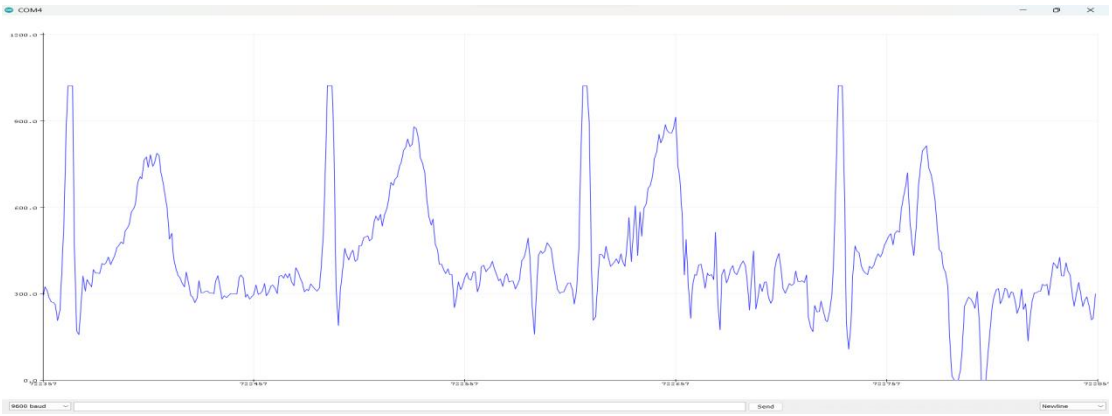


Figure 3 : ECG signal readings of the patient

As seen in figure 3, the ECG signal of the patient from Image 2 is really clean and smooth and the serial values of the serial plotter shows that the ECG pulse is between 300 to above 900 *(presumably around 1000)* proving that the filtration and amplification of the circuit works accordingly in an excellent manner. However, when observing the PQRST wave of the ECG signal, there is a noticeable saturation on the R wave meaning that the amplification is too high where the serial values are above

1023, the maximum value of the Arduino Uno. In spite of these shortcomings, the overall experiment was still a success.

Discussion and Conclusion

The experiment that was done was done on a breadboard with long wires and large components. In the modern age, most medical devices are now worn like the Apple smartwatch or the AliveCor KardiaBand (KB) smartwatch that applies the concept of ECG to detect atrial fibrillation in the users' hearts*[Rajakariar et al, 2020][Isakadze and Marin, 2019]*. Thus, the practical approach to a real life application such as the smartwatch is to build the ECG circuit on a smaller circuit interface with smaller electrical components.

Another thing to consider is that the experimental circuit used is a single supply so as to avoid using the power supply that is connected to the mains supply. The reason this risk is taken into precaution is due to the absence of shielding. In many heavy electrical-related industry like a hospital that uses a lot of power supply for the medical operations used in the building, the mains supply is isolated from the medical device used keeping the staffs and patients safe from any hazards and risks like an electrical shock or a short circuit.*[Portal and Edvard, 2021]*. In conclusion, the isolation of the mains supply from the ECG must be applied in order for the experiment to be integrated into a more realistic usage of ECGs.

REFERENCES

Abi-Saleh, B. and Omar, B. (2010). *Einthoven's Triangle Transparency: A Practical Method to Explain Limb Lead Configuration Following Single Lead Misplacements. Reviews in Cardiovascular Medicine*, 11(1), pp.33–38. doi:10.3909/ricm0506.

Christov, I., Neycheva, T., Schmid, R., Stoyanov, T. and Abächerli, R. (4 February 2017). *Pseudo-real-time low-pass filter in ECG, self-adjustable to the frequency spectra of the waves. Medical & Biological Engineering & Computing*, 55(9), pp.1579–1588. doi:10.1007/s11517-017-1625-y.

hades.mech.northwestern.edu. (14 December 2008). *Second Order Active Filters - Northwestern Mechatronics Wiki*. [online] Available at: http://hades.mech.northwestern.edu/index.php/Second_Order_Active_Filters#:~:text=2nd%20order%20active%20filtering%20has,impedance%20input%2C%20low%20impedance%20output [Accessed 28 Dec. 2022].

Isakadze, N. and Martin, S.S. (2019). *How useful is the smartwatch ECG? Trends in Cardiovascular Medicine*, [online] 30(7). doi:10.1016/j.tcm.2019.10.010.

Luca, C.J. (2002). *Surface electromyography: Detection and recording*. [online] www.semanticscholar.org. Available at: https://www.semanticscholar.org/paper/Surface-electromyography%3A-Detection-and-recording-Luca/992836d36c1ae4b56314e8b15f087822a1995c93 [Accessed 5 Jan. 2023]

Mayo Clinic (2022). *Electrocardiogram (ECG or EKG) - Mayo Clinic*. [online] www.mayoclinic.org. Available at: https://www.mayoclinic.org/tests-procedures/ekg/about/pac-20384983#:~:text=An%20electrocardiogram%20(ECG%20or%20EKG,attached%20computer%20monitor%20or%20printer.

nhs.uk. (2017). *Electrocardiogram (ECG)*. [online] Available at: https://www.nhs.uk/conditions/electrocardiogram/#:~:text=Generally%2C%20the%20test%20involves%20attaching,and%20drink%20as%20normal%20beforehand.

Kher, R. (12 March 2019). *Signal Processing Techniques for Removing Noise from ECG Signals. J Biomed Eng Res*, [online] 3, p.101. Available at: http://www.jscholaronline.org/articles/JBER/Signal-Processing.pdf

Pizzuti, G.P., Cifaldi, S. and Nolfe, G. (1985). *Digital sampling rate and ECG analysis. Journal of Biomedical Engineering*, 7(3), pp.247–250. doi:10.1016/0141-5425(85)90027-5.

Portal, E.-E.E. and Edvard (2021). *Isolation power solutions for ultimate power availability in hospitals | EEP. [online] EEP - Electrical Engineering Portal*. Available at: https://electrical-engineering-portal.com/isolation-power-solutions-hospital-power-availability-efficiency-safety [Accessed 28 Dec. 2022].

Rajakariar, K., Koshy, A.N., Sajeev, J.K., Nair, S., Roberts, L. and Teh, A.W. (7 January 2020). *Accuracy of a smartwatch based single-lead electrocardiogram device in detection of atrial fibrillation*. Heart, p.heartjnl-2019-316004. doi:10.1136/heartjnl-2019-316004.

Staff, E. (2012). *Interrupts short and simple: Part 1 - Good programming practices*. [online] Embedded.com. Available at: https://www.embedded.com/interrupts-short-and-simple-part-1-good-programming-practices/ [Accessed 7 Jan. 2023].

Texas Instruments, (2022), INAx126 MicroPower Instrumentation Amplifiers[online] Available at : https://www.ti.com/lit/ds/symlink/ina126.pdf

www.cardiosecur.com. (n.d.). *Lead systems – how an ECG works | CardioSecur*. [online] Available at: https://www.cardiosecur.com/magazine/specialist-articles-on-the-heart/lead-systems-how-an-ecg-works.

www.learningaboutelectronics.com. (n.d.). *What is the Difference Between a Single and Dual Supply Op Amp?* [online] Available at: http://www.learningaboutelectronics.com/Articles/Difference-between-a-single-and-dual-supply-op-amp.php .

Xiu,L and Li,Z, (2012), *Low-Power Instrumentation Amplifier IC Design for ECG System Applications*, Procedia Engineering 29 (2012, p1533-1538, Available online at : https://www.researchgate.net/publication/271891199_Low-Power_Instrumentation_Amplifier_IC_Design_for_ECG_System_Applications

Zahak, M. (2012). *Signal Acquisition Using Surface EMG and Circuit Design Considerations for Robotic Prosthesis. Computational Intelligence in Electromyography Analysis - A Perspective on Current Applications and Future Challenges*. [online] doi:10.5772/52556

APPENDIX
Whole Circuit for ECG

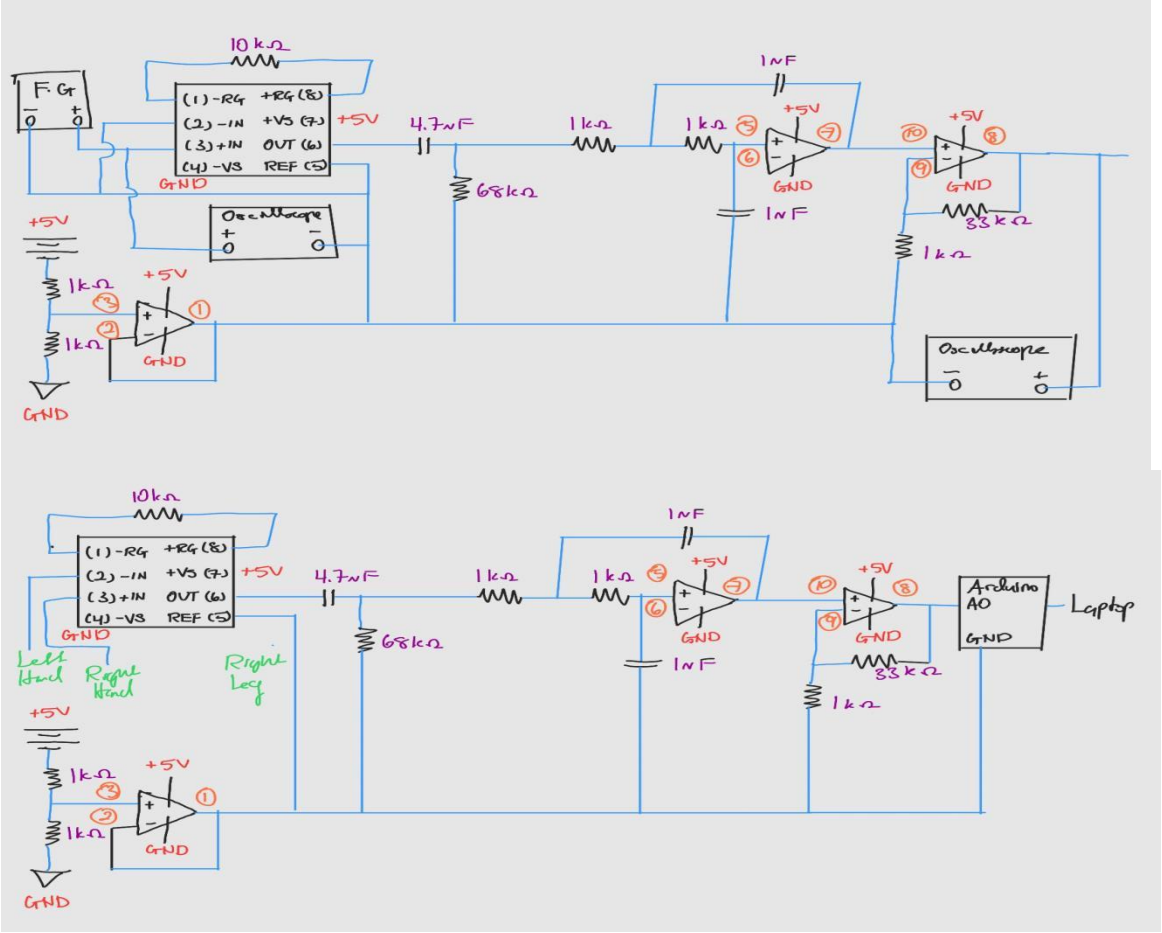


Image 3 : Whole Circuit for the test run(top) and for the electrodes(bottom)

Tabled Data for High Pass Filter

HIGH PASS FILTER			
Frequency(Hz)	Vout(V)	Vin(V)	Gain(dB)
0.05	0.528	1.02	-5.719324985
0.1	0.568	1.02	-5.085036721
0.15	0.568	1.02	-5.085036721
0.2	0.6	1.02	-4.608978428
0.25	0.656	1.02	-3.833926648
0.3	0.648	1.02	-3.940503318
0.35	0.656	1.01	-3.748350688

0.4	0.672	1.02	-3.624617974
0.45	0.648	1.02	-3.940503318
0.5	0.686	1.01	-3.359945162
0.55	0.696	1.01	-3.234242683
0.6	0.728	1.02	-2.929375849
0.65	0.736	1.01	-2.748871189
0.7	0.752	1.01	-2.562070664
0.75	0.768	1.01	-2.379203075
0.8	0.8	1.01	-2.024627736
0.85	0.8	1	-1.93820026
0.9	0.8	1	-1.93820026
0.95	0.8	1	-1.93820026
1	0.8	1.01	-2.024627736

Tabled Data for 2nd Order Sallen Key Low Pass Filter

2ND ORDER LOW PASS FILTER			
Frequency(Hz)	Vout(V)	Vin(V)	Gain(dB)
20	0.984	1	-0.140098031
40	0.904	1	-0.87663139
60	0.812	0.992	-1.739112858
70	0.76	0.976	-2.172724508
80	0.712	0.976	-2.739396481
90	0.664	0.976	-3.345634766
100	0.616	0.976	-3.99738211
110	0.584	0.968	-4.389250204
120	0.544	0.976	-5.077018359
130	0.504	0.976	-5.740385624
140	0.472	0.976	-6.310156381
150	0.448	0.976	-6.763436073
160	0.416	0.976	-7.407129741
170	0.392	0.976	-7.923275013
180	0.360	0.976	-8.662946338
190	0.344	0.968	-8.986338295
200	0.312	0.976	-9.905904473
210	0.3	0.976	-10.24657126

Microcontroller Programming for ECG experiment via Arduino IDE

```
static inline void initTimer (void)
{
  //write code to initialize TIMER 1
  TCNT1 = 65411;                              //PRELOAD TIMER OF TIMER 1
  TCCR1B |= (1 << CS12);                    //256 PRESCALER FOR SINCE CS12 IS SET TO 1 AND CS11
AND CS10 IS 0 BY DEFAULT
  TIMSK1 |= (1 << TOIE1);                   // ENABLE TIMER OVERFLOW INTERRUPT AS TOIE1 IS SET T
O 1 WHICH ALLOWS THE ISR OVERFLOW CODE TO RUN
}
```

```
static inline void initIOPorts (void)
{
  // ANALOG A0 IS THE INPUT PIN BY DEFAULT SINCE MUX0,MUX1,MUX2 AND MUX3 IS SET TO 0 BY DEFAUL
T
}
```

```
static inline void initADC (void)
{
  ADMUX |= (1 << REFS0);                                // SET THE Vref TO AREF PIN TO SET TH
E REFFERNCE VOLTAGE TO 5V WHEN THE CONVERSION STARTS
  ADCSRA |= (1 << ADPS2);                               // SET THE 16-BIT PRESCALER
  ADCSRA |= (1 << ADEN);                                // ENABLE ADC CONVERSION
  ADCSRA |= (1 << ADIE);                                // ENABLE ADC INTERRUPT AS THE ADIE(A
DC INTERRUPT ENABLE) IS SET TO 1, ACTIVATING THE ADC CONVERSION COMPLETE INTERRUPT
}
```

```
//----INTERRUPT SERVICE ROUTINE----//
ISR (TIMER1_OVF_vect)                   // ONCE THE TIMER OVERFLOWS, IT WILL EXECUTE THE ISR OV
ERFLOW COMMAND WHICH STARTS ADC CONVERSION
{
  ADCSRA |= (1 << ADSC);
  TCNT1 = 65411;       // BEGINS THE ADC CONVERSION
}
```

```
ISR (ADC_vect)                          // ONCE A CONVERSION IS DONE, THE ISR ADC INTERRUPT COM
MAND WILL BE EXECUTED WHICH PRINTS ADC VALUE AT BAUD RATE 9600
{
  Serial.begin(9600);
  Serial.println (ADC);
```

```
}
```

```
int main (void)
{
  noInterrupts ();
  initIOPorts ();
  initTimer ();
  initADC ();
  interrupts ();
```

```
  while (1)
  {
```

```
  }
  return 0;
}
```